Related Rates

ANSWER KEY

Section 1: Geometric Expansion (Circles, Spheres, Cubes)

1 The radius of a circle expands at a constant rate of 3 cm/s. How fast is the area increasing when the radius is 10 cm?

a. 60π cm^2/s ✓

b. 30π cm^2/s

c. 100π cm^2/s

d. 20π cm^2/s

$A = \pi r^2 \Rightarrow \frac{dA}{dt} = 2\pi r\frac{dr}{dt} = 2\pi(10)(3) = 60\pi$ cm^2/s.

2 Air is pumped into a spherical balloon so that its volume increases at 100 cm^3/s. How fast is the radius increasing when $r = 5$ cm?

a. $\frac{1}{\pi}$ cm/s ✓

b. $\frac{2}{\pi}$ cm/s

c. $\frac{4}{\pi}$ cm/s

d. 5π cm/s

$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2\frac{dr}{dt}$. $100 = 4\pi(25)\frac{dr}{dt} = 100\pi\frac{dr}{dt} \Rightarrow \frac{dr}{dt} = \frac{1}{\pi}$ cm/s.

3 The edge length s of a cube increases at 2 cm/s. At what rate is the surface area increasing when $s = 4$ cm?

a. 96 cm^2/s ✓

b. 48 cm^2/s

c. 32 cm^2/s

d. 64 cm^2/s

Surface area $S = 6s^2 \Rightarrow \frac{dS}{dt} = 12s\frac{ds}{dt} = 12(4)(2) = 96$ cm^2/s.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Section 2: Sliding Ladders and Pythagorean Triangles

4 A 13-ft ladder leans against a vertical wall. If the bottom slides away from the wall at 2 ft/s, how fast is the top sliding down when the bottom is 5 ft from the wall?

a. 5/6 ft/s ✓

b. 12/5 ft/s

c. 2 ft/s

d. 1/2 ft/s

$x^2 + y^2 = 13^2 = 169$. When $x = 5$, $y = \sqrt{169 - 25} = 12$. Differentiating:
$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \Rightarrow 5(2) + 12\frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = -\frac{10}{12} = -\frac{5}{6}$ ft/s.

5 Two cars start from the same point. Car A travels north at 40 mph and Car B travels east at 30 mph. How fast is the distance between them increasing after 2 hours?

a. 50 mph ✓

b. 70 mph

c. 45 mph

d. 60 mph

After 2 h: $y = 80$ miles, $x = 60$ miles, $z = \sqrt{60^2 + 80^2} = 100$ miles.
$2z\frac{dz}{dt} = 2x\frac{dx}{dt} + 2y\frac{dy}{dt} \Rightarrow 100\frac{dz}{dt} = 60(30) + 80(40) = 1800 + 3200 = 5000 \Rightarrow \frac{dz}{dt} = 50$ mph.

6 In the 13-ft ladder problem ($x = 5, y = 12, dx/dt = 2$), at what rate is the area of the triangle formed by the ladder changing?

a. 119/12 ft^2/s ✓

b. 25/2 ft^2/s

c. 0 ft^2/s

d. 12 ft^2/s

$A = \frac{1}{2}xy \Rightarrow \frac{dA}{dt} = \frac{1}{2}\left(\frac{dx}{dt}y + x\frac{dy}{dt}\right) = \frac{1}{2}(2(12) + 5(-5/6)) = \frac{1}{2}(24 - 25/6) = \frac{1}{2}\left(\frac{119}{6}\right) = \frac{119}{12}$ ft^2/s.

Section 3: Conical and Cylindrical Tanks

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 Water flows into an inverted conical tank (radius 4 m, height 12 m) at 2 m^3/min. How fast is the water level rising when the depth is 6 m?

a. $\frac{1}{2\pi}$ m/min ✓

b. $\frac{1}{\pi}$ m/min

c. $\frac{2}{\pi}$ m/min

d. $\frac{1}{4\pi}$ m/min

By similar triangles, $r/h = 4/12 = 1/3 \Rightarrow r = h/3$. $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(h/3)^2 h = \frac{\pi}{27}h^3$. $\frac{dV}{dt} = \frac{\pi}{9}h^2\frac{dh}{dt}$. At $h = 6$: $2 = \frac{\pi}{9}(36)\frac{dh}{dt} = 4\pi\frac{dh}{dt} \Rightarrow \frac{dh}{dt} = \frac{2}{4\pi} = \frac{1}{2\pi}$ m/min.

8 A cylindrical tank of radius 3 m is being filled at 18π m^3/min. How fast is the water height increasing?

a. 2 m/min ✓

b. 3 m/min

c. 6 m/min

d. 1 m/min

$V = \pi r^2 h = \pi(9)h = 9\pi h$. $\frac{dV}{dt} = 9\pi\frac{dh}{dt}$. $18\pi = 9\pi\frac{dh}{dt} \Rightarrow \frac{dh}{dt} = 2$ m/min.

9 A trough with triangular cross-section (base 2 m, depth 2 m, length 6 m) leaks water at 0.5 m^3/min. What is dh/dt when $h = 1$ m?

a. $-1/12$ m/min ✓

b. $-1/6$ m/min

c. $-1/8$ m/min

d. $-1/4$ m/min

Cross section base $b = h$. $V = \frac{1}{2}bh \times 6 = 3h^2$. $\frac{dV}{dt} = 6h\frac{dh}{dt}$. $-0.5 = 6(1)\frac{dh}{dt} \Rightarrow \frac{dh}{dt} = -\frac{0.5}{6} = -\frac{1}{12}$ m/min.

Section 4: Shadows and Angles of Elevation

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 A 6-ft tall person walks away from a 15-ft streetlight at 4 ft/s. At what rate is the length of the person's shadow increasing?

a. 8/3 ft/s ✓

b. 2 ft/s

c. 4 ft/s

d. 10/3 ft/s

Similar triangles: $\frac{s}{6} = \frac{x+s}{15} \Rightarrow 15s = 6x + 6s \Rightarrow 9s = 6x \Rightarrow s = \frac{2}{3}x$. $\frac{ds}{dt} = \frac{2}{3}\frac{dx}{dt} = \frac{2}{3}(4) = \frac{8}{3}$ ft/s.

11 In the same streetlight scenario, at what rate is the tip of the shadow moving?

a. 20/3 ft/s ✓

b. 8/3 ft/s

c. 4 ft/s

d. 12 ft/s

Tip position is $x + s = x + \frac{2}{3}x = \frac{5}{3}x$. Rate of tip $= \frac{5}{3}\frac{dx}{dt} = \frac{5}{3}(4) = \frac{20}{3}$ ft/s.

12 An observer 100 m from a rocket launch site watches a rocket rise vertically at 50 m/s. Find $\frac{d\theta}{dt}$ when the rocket is 100 m high.

a. 0.25 rad/s ✓

b. 0.50 rad/s

c. 1.00 rad/s

d. 0.10 rad/s

$\tan\theta = \frac{y}{100}$. $\sec^2\theta\frac{d\theta}{dt} = \frac{1}{100}\frac{dy}{dt}$. When $y = 100$, $\theta = \pi/4$, so $\sec^2(\pi/4) = 2$.
$2\frac{d\theta}{dt} = \frac{1}{100}(50) = 0.5 \Rightarrow \frac{d\theta}{dt} = 0.25$ rad/s.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647